

PATENT RESEARCH REPORT

1. ABSTRACT

This technical report synthesizes polymerization and process parameters for Low Acrylonitrile Nitrile Butadiene Rubber (Low ACN NBR) based on a selected portfolio of six patents published between 2011 and 2017. Low ACN NBR (typically containing 27% to 36% bound acrylonitrile) is highly valued for its low-temperature flexibility and oil resistance. The analyzed patent landscape demonstrates that cold emulsion polymerization conducted at 7% to 8°C is the dominant industrial approach to synthesize these copolymers. Key process trends include the use of advanced redox initiator systems (such as diisopropylbenzene hydroperoxide or p-menthane hydroperoxide coupled with iron(II) salts and reducing agents like sodium formaldehyde sulfoxylate or Rongalit C) to achieve high conversion (70% to 90%) within 6 to 7 hours. Molecular weight and Mooney viscosity are tightly regulated using tertiary dodecyl mercaptan (t-DDM) or tert-dodecyl mercaptan (TDM), often employing incremental booster additions during the reaction. Furthermore, post-polymerization coagulation conditions—specifically the choice of salt (MgCl₂, CaCl₂, MgSO₄, or NaCl) and precise pH adjustment prior to coagulation—are shown to be critical in controlling the polymer's ion index, which directly dictates vulcanization kinetics, Mooney scorch safety, and final mechanical properties.

2. METHODOLOGY

POLYMERIZATION RECIPE EXTRACTIONS

Patent 1

Patent Details - Patent Number: US9650502 - Patent Title: Process for the preparation of nitrile rubbers - Assignee: Not disclosed in the analyzed patent evidence. - Jurisdiction: US - Publication Year: 2017 - Polymer Type: Nitrile Butadiene Rubber (NBR) - Relevance to target: Discloses a low-temperature emulsion polymerization process for NBR with low-to-medium bound ACN content (27.9% to 33.9%) utilizing incremental chain-transfer agent additions.

Polymerization / Synthesis Method - Polymerization process: Cold emulsion polymerization. - Monomer system: Acrylonitrile and butadiene. - Monomer ratio: Latex A: 500 g acrylonitrile and 4.37 kg butadiene (~10.3% ACN in feed). Latex B: 815 g acrylonitrile and 4.0 kg butadiene (~16.9% ACN in feed). - Water amount: 9.92 kg of deionized water. - Emulsifier: Potassium stearate and potassium salt of disproportionated rosin acid (DNMK). - Emulsifier loading: 0.53 kg of 5.4% wt aqueous potassium stearate solution and 0.440 kg of DNMK. - Initiator: Diisopropylbenzene hydroperoxide (DIHP). - Initiator loading: 6 g. - Catalyst / activator: Sodium formaldehyde sulfoxylate (FSS) (139 g of aqueous solution). - Chain-transfer agent: Tert-dodecyl

mercaptan (TDM). - Chain-transfer dosage: 18.72 g initially (for Latex A), with subsequent booster additions of 2.34 g at 37% and 59% conversion. - Polymerization temperature: 7°C. - Pressure: Not disclosed in the analyzed patent evidence. - pH: 8.5 to 8.6 (latex pH before coagulation); coagulation pH adjusted to 11.5-11.8. - Reaction time: 7 hours. - Conversion: 70% conversion. - Coagulation conditions: Latex (10 kg) added over 10 minutes to a mixture of 43 kg water and 4 kg saturated magnesium sulfate solution at 45°C, with pH adjusted to 11.5-11.8 using potassium hydroxide (14 g). - Post-treatment: Crumbs (1.5 kg) filtered, washed with deionized water, and dried in an air oven at 100°C for 24 hours to a residual humidity of less than 1%. - Raw polymer properties: Latex A: Bound ACN 28.1%, Mooney Viscosity 60.5 MU. Latex B: Bound ACN 33.9%, Mooney Viscosity 32.1 MU. Latex C: Bound ACN 27.9%, Mooney Viscosity 29.0 MU.

Relevant Experimental Evidence - Latex A was prepared by charging 9.92 kg deionized water, potassium stearate, and DNMK, followed by vacuum application and charging 500 g ACN, 18.72 g TDM, and 4.37 kg butadiene. - Polymerization was initiated at 7°C by injecting 6 g DIHP and 139 g FSS solution. - TDM boosters (2.34 g each) were added at 37% and 59% conversion to control molecular weight. - Reaction was shortstopped at 70% conversion using 515 g of 2% wt hydroxylamine sulfate solution. - Latex was stabilized with 1% wt BHT antioxidant dispersion before being coagulated with magnesium sulfate at pH 11.5-11.8.

Technical Relevance Demonstrates that split-addition of TDM chain transfer agent during cold emulsion polymerization successfully controls Mooney viscosity in low-ACN nitrile rubbers.

Patent 2

Patent Details - Patent Number: US9034992 - Patent Title: Nitrile rubbers which optionally contain alkylthio terminal groups and which ... - Assignee: Not disclosed in the analyzed patent evidence. - Jurisdiction: US - Publication Year: 2015 - Polymer Type: Nitrile Butadiene Rubber (NBR) - Relevance to target: Focuses on the coagulation and vulcanization of NBR latices to optimize mechanical and aging properties.

Polymerization / Synthesis Method - Polymerization process: Emulsion polymerization. - Monomer system: Acrylonitrile and butadiene. - Monomer ratio: Not disclosed in the analyzed patent evidence. - Water amount: Not disclosed in the analyzed patent evidence. - Emulsifier: Not disclosed in the analyzed patent evidence. - Emulsifier loading: Not disclosed in the analyzed patent evidence. - Initiator: Not disclosed in the analyzed patent evidence. - Initiator loading: Not disclosed in the analyzed patent evidence. - Catalyst / activator: Not disclosed in the analyzed patent evidence. - Chain-transfer agent: Not disclosed in the analyzed patent evidence. - Chain-transfer dosage: Not disclosed in the analyzed patent evidence. - Polymerization temperature: Not disclosed in the analyzed patent evidence. - Pressure: 120 bar (during vulcanization/pressing). - pH: Not disclosed in the analyzed patent evidence. - Reaction time: Not disclosed in the analyzed patent evidence. - Conversion: 90% conversion (for vulcanization testing). - Coagulation conditions: Coagulated using magnesium chloride in the presence of an acid-ashed gelatin (viscosity in 10% aq solution: 92.1 cP; isoelectric point: 8.7). - Post-treatment: Vulcanization at 160°C under 120 bar hydraulic pressure for 30 minutes. - Raw polymer properties: Mooney sheets subjected to hot air aging at 100°C for 48 hours; vulcanizates

evaluated for stress at 300% elongation, tensile strength, and elongation at break.

Relevant Experimental Evidence - Nitrile rubber mixtures were prepared in a 1.5 L laboratory internal mixer. - Latex coagulation was performed using 25 L of Latex A with magnesium chloride and an acid-ashed gelatin co-agent. - Vulcanized specimens were pressed at 120 bar and 160°C for 30 minutes. - Corrosion testing of sealing rings was conducted at 80% relative humidity and 70°C between aluminum plates.

Technical Relevance Highlights the use of magnesium chloride and gelatin during latex coagulation to improve the vulcanizate properties and corrosion resistance of nitrile rubbers.

Patent 3

Patent Details - Patent Number: US7923518 - Patent Title: Nitrile rubbers - Assignee: Not disclosed in the analyzed patent evidence. - Jurisdiction: US - Publication Year: 2011 - Polymer Type: Nitrile Butadiene Rubber (NBR) - Relevance to target: Relates to the synthesis and processing of nitrile rubbers.

Polymerization / Synthesis Method - Polymerization process: Emulsion polymerization. - Monomer system: Acrylonitrile and butadiene. - Monomer ratio: Not disclosed in the analyzed patent evidence. - Water amount: Not disclosed in the analyzed patent evidence. - Emulsifier: Not disclosed in the analyzed patent evidence. - Emulsifier loading: Not disclosed in the analyzed patent evidence. - Initiator: Not disclosed in the analyzed patent evidence. - Initiator loading: Not disclosed in the analyzed patent evidence. - Catalyst / activator: Not disclosed in the analyzed patent evidence. - Chain-transfer agent: Not disclosed in the analyzed patent evidence. - Chain-transfer dosage: Not disclosed in the analyzed patent evidence. - Polymerization temperature: Not disclosed in the analyzed patent evidence. - Pressure: Not disclosed in the analyzed patent evidence. - pH: Not disclosed in the analyzed patent evidence. - Reaction time: Not disclosed in the analyzed patent evidence. - Conversion: Not disclosed in the analyzed patent evidence. - Coagulation conditions: Not disclosed in the analyzed patent evidence. - Post-treatment: Not disclosed in the analyzed patent evidence. - Raw polymer properties: Not disclosed in the analyzed patent evidence.

Relevant Experimental Evidence - No specific experimental parameters were disclosed in the analyzed patent text.

Technical Relevance Provides baseline patent context for the development of nitrile rubber compositions.

Patent 4

Patent Details - Patent Number: US8389623 - Patent Title: Nitrile rubbers - Assignee: Not disclosed in the analyzed patent evidence. - Jurisdiction: US - Publication Year: 2013 - Polymer Type: Nitrile Butadiene Rubber (NBR) - Relevance to target: Investigates the impact of coagulation salts on the vulcanization kinetics and mechanical properties of NBR.

Polymerization / Synthesis Method - Polymerization process: Emulsion polymerization. - Monomer system: Acrylonitrile and butadiene. - Monomer ratio: Not disclosed in the analyzed patent evidence. - Water amount: Not disclosed in the analyzed patent evidence. - Emulsifier: Not disclosed in the analyzed patent evidence. - Emulsifier loading: Not disclosed in the analyzed patent evidence. - Initiator: Not disclosed in the analyzed patent evidence. - Initiator loading: Not disclosed in the analyzed patent evidence. - Catalyst / activator: Not disclosed in the analyzed patent evidence. - Chain-transfer agent: Not disclosed in the analyzed patent evidence. - Chain-transfer dosage: Not disclosed in the analyzed patent evidence. - Polymerization temperature: Not disclosed in the analyzed patent evidence. - Pressure: 120 bar (during vulcanization). - pH: Not disclosed in the analyzed patent evidence. - Reaction time: Not disclosed in the analyzed patent evidence. - Conversion: 90% conversion (for vulcanization testing). - Coagulation conditions: Coagulated using magnesium chloride or calcium chloride. - Post-treatment: Vulcanization at 160°C under 120 bar hydraulic pressure for 30 minutes. - Raw polymer properties: Evaluated for Mooney scorch, vulcanization rate (t90 - t10), and stress at 300% elongation.

Relevant Experimental Evidence - Nitrile rubbers were processed in a 1.5 L mechanical laboratory mixer with compounding ingredients reported in parts per 100 parts of rubber. - Vulcanization kinetics were measured according to DIN 53 529, part 3, identifying t10 and t90 times. - Coagulation with magnesium chloride resulted in lower Mooney scorch, shorter vulcanization times, and higher stress at 300% elongation compared to calcium chloride coagulation.

Technical Relevance Establishes that magnesium chloride coagulation optimizes vulcanization rates and enhances tensile stress at high elongation for NBR.

Patent 5

Patent Details - Patent Number: US8436091 - Patent Title: Nitrile rubbers - Assignee: Not disclosed in the analyzed patent evidence. - Jurisdiction: US - Publication Year: 2013 - Polymer Type: Nitrile Butadiene Rubber (NBR) - Relevance to target: Discloses a highly detailed emulsion polymerization recipe for nitrile rubbers containing low-to-medium ACN (35-36%) using specific redox catalyst systems.

Polymerization / Synthesis Method - Polymerization process: Cold emulsion polymerization. - Monomer system: Butadiene, acrylonitrile, and 2-hydroxyethyl methacrylate. - Monomer ratio: 61.4 parts butadiene, 36 parts acrylonitrile, and 2.6 parts 2-hydroxyethyl methacrylate. - Water amount: 220 parts by weight per 100 parts monomer (total water in autoclave: 2.1 kg). - Emulsifier: Sodium lauryl sulphate (Texapon K-12). - Emulsifier loading: 3.0 parts by weight per 100 parts monomer. - Initiator: P-menthane hydroperoxide (Trigonox NT 50) and Fe(II)SO₄·7H₂O or Fe(II)Cl₂·4H₂O. - Initiator loading: 0.02 parts by weight Trigonox NT 50; 0.030 parts by weight Fe(II) salt premix. - Catalyst / activator: Rongalit C (2.0 g) and EDTA in an equimolar amount relative to Fe(II). - Chain-transfer agent: Tert-dodecyl mercaptan (t-DDM). - Chain-transfer dosage: 0.54 parts by weight per 100 parts monomer. - Polymerization temperature: 8.0 % 0.5°C. - Pressure: 50 bar to 150 bar (for subsequent hydrogenation steps, if applicable). - pH: 7.0 % 1.0. - Reaction time: 6.5

to 7.0 hours. - Conversion: 84.2% to 86.1%. - Coagulation conditions: Latex admixed with a 50% dispersion of Vulkanox BKF (0.3% wt based on NBR solids) prior to coagulation. - Post-treatment: Shortstopped with diethylhydroxylamine (0.2 parts), stabilized with Wingstay 29/Naugawhite antioxidant mixture, and coagulated. - Raw polymer properties: ACN content: 35.3% - 36.6%. Mooney viscosity: 25 - 142 MU. Glass transition temperature (Tg): -23.1°C to -25.8°C.

Relevant Experimental Evidence - Autoclave batches utilized 1.25 kg monomer mixture and 2.1 kg water. - Premix solution contained 0.986 g Fe(II)SO₄·7H₂O or 0.845 g Fe(II)Cl₂·4H₂O and 2.0 g Rongalit C in 400 g water. - Polymerization of Rubber G (FeSO₄ premix) reached 86.1% conversion in 7.0 hours; Rubber H (FeCl₂ premix) reached 84.2% conversion in 6.5 hours. - Antioxidant stabilization was performed using a mixture of Sorbilene Mix (25 g), Naugawhite (38 g), Wingstay 29 (125 g), and water (63 g).

Technical Relevance Provides a precise, reproducible cold emulsion polymerization recipe demonstrating that specific Fe(II) redox systems control branching and Mooney viscosity in low-to-medium ACN rubbers.

Patent 6

Patent Details - Patent Number: US8664340 - Patent Title: Nitrile rubbers - Assignee: Not disclosed in the analyzed patent evidence. - Jurisdiction: US - Publication Year: 2014 - Polymer Type: Nitrile Butadiene Rubber (NBR) - Relevance to target: Investigates the effect of latex pH adjustment and salt selection during coagulation on the vulcanization rate of NBR.

Polymerization / Synthesis Method - Polymerization process: Emulsion polymerization. - Monomer system: Acrylonitrile and butadiene. - Monomer ratio: Not disclosed in the analyzed patent evidence. - Water amount: Not disclosed in the analyzed patent evidence. - Emulsifier: Not disclosed in the analyzed patent evidence. - Emulsifier loading: Not disclosed in the analyzed patent evidence. - Initiator: Not disclosed in the analyzed patent evidence. - Initiator loading: Not disclosed in the analyzed patent evidence. - Catalyst / activator: Not disclosed in the analyzed patent evidence. - Chain-transfer agent: Not disclosed in the analyzed patent evidence. - Chain-transfer dosage: Not disclosed in the analyzed patent evidence. - Polymerization temperature: Not disclosed in the analyzed patent evidence. - Pressure: Not disclosed in the analyzed patent evidence. - pH: Latex pH adjusted to values ranging from 5.8 to 9.5 prior to coagulation. - Reaction time: Not disclosed in the analyzed patent evidence. - Conversion: Not disclosed in the analyzed patent evidence. - Coagulation conditions: Coagulated using NaCl or CaCl₂. Latex pH adjusted using 5% strength aqueous KOH or HCl before coagulation. - Post-treatment: Vulcanization testing of unvulcanized rubber mixtures prepared in a 1.5 L laboratory mixer. - Raw polymer properties: Evaluated for ion index, vulcanization rate (t₉₀ - t₁₀), and stress at 30% elongation.

Relevant Experimental Evidence - Latex pH was adjusted to 9.5 or 8.4 using 5% KOH, or to 5.8 using 5% HCl before coagulation. - Coagulation trials compared NaCl (15-26% wt) and CaCl₂ (0.3-1.2% wt) at temperatures from 20°C to 50°C. - Nitrile rubbers with optimized ion indices exhibited significantly higher vulcanization rates (t₉₀ - t₁₀) and higher stress values at 30%

elongation.

Technical Relevance Demonstrates that pre-coagulation pH adjustment and salt selection are critical process controls to regulate the ion index and enhance vulcanization kinetics.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- No cross-patent comparison provided.

4. REFERENCES

#	Patent No.	Title	Assignee	Jurisdiction	Year	Source
1	US7923518	Nitrile rubbers	Not disclosed	US	2011-04-12	Google Patents
2	US8389623	Nitrile rubbers	Not disclosed	US	2013-03-05	Google Patents
3	US8436091	Nitrile rubbers	Not disclosed	US	2013-05-07	Google Patents
4	US8664340	Nitrile rubbers	Not disclosed	US	2014-03-04	Google Patents
5	US9034992	Nitrile rubbers which optionally contain alkylthio terminal groups and which ...	Not disclosed	US	2015-05-19	Google Patents

6	US9650502	Process for the preparation of nitrile rubbers	Not disclosed	US	2017-05-16	Google Patents
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